



Standard Form for P.U.L RLGC Matrices of MTLs

Theory, Mathematical Foundations, Complete Stamp Catalog, and Simulation Algorithms

- The **per-unit-length inductance matrix**, $\mathbf{L}$, contains the individual per-unit-length self-inductances, l_{ii} , of the circuits and the per-unit-length mutual inductance between the circuits, l_{ij} , as

$$\mathbf{L} = \begin{bmatrix} l_{11} & l_{12} & \cdots & l_{1n} \\ l_{21} & l_{22} & \cdots & l_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ l_{n1} & l_{n2} & \cdots & l_{nn} \end{bmatrix} \quad (1)$$

- Similarly, we define the **per-unit-length resistance matrix** as

$$\mathbf{R} = \begin{bmatrix} (r_1 + r_0) & r_0 & \cdots & r_0 \\ r_0 & (r_2 + r_0) & \cdots & r_0 \\ \vdots & \vdots & \ddots & \vdots \\ r_0 & r_0 & \cdots & (r_n + r_0) \end{bmatrix} \quad (2)$$

where $\mathbf{V}$ and $\mathbf{I}$ are defined by (2.11).

- The **per-unit-length conductance matrix**, $\mathbf{G}$, represents the conduction current flowing between the conductors in the transverse plane as

$$\mathbf{G} = \begin{bmatrix} \sum_{k=1}^n g_{1k} & -g_{12} & \cdots & -g_{1n} \\ -g_{21} & \sum_{k=1}^n g_{2k} & \cdots & -g_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -g_{n1} & -g_{n2} & \cdots & \sum_{k=1}^n g_{nk} \end{bmatrix} \quad (3)$$

- The **per-unit-length capacitance matrix**, $\mathbf{C}$, represents the displacement current flowing between the conductors in the transverse plane as

$$\mathbf{C} = \begin{bmatrix} \sum_{k=1}^n c_{1k} & -c_{12} & \cdots & -c_{1n} \\ -c_{21} & \sum_{k=1}^n c_{2k} & \cdots & -c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -c_{n1} & -c_{n2} & \cdots & \sum_{k=1}^n c_{nk} \end{bmatrix} \quad (4)$$

References

- [1] C. R. Paul, *Analysis of Multiconductor Transmission Lines*. New York, NY, USA: John Wiley & Sons, Inc., 1994, ch. 2, pp. 51–55.